Example 1: Streaming Time

Sample A ($n_1 = 14$): $\bar{x}_1 = 52$ min, $s_1 = 4.1$. Sample B ($n_2 =$

10): $\bar{x}_2 = 48$ min, $s_2 = 6.3$. Test $\alpha = 0.05$, two-tailed.

- $t = (52 - 48)/\sqrt{(4.1^2/14 + 6.3^2/10)} \approx 1.80$.
- $v \approx 16.3$, $t^* \approx \pm 2.12 \Rightarrow |t| < t^* \Rightarrow$ fail to reject H_0.
- 95% CI $\approx (-0.9, 8.9)$ minutes.

Example 2: Assembly Output

Line A ($n_1 = 9$): $\bar{x}_1 = 105$, $s_1 = 5$. Line B ($n_2 = 13$): $\bar{x}_2 =$

98, $s_2 = 3$. Test $\alpha = 0.01$, right-tailed.

- $t = (105 - 98)/\sqrt{(5^2/9 + 3^2/13)} \approx 4.15$.
- $v \approx 12.5$, one-tailed critical $t \approx 2.68 \Rightarrow$ reject H_0.
- 99% CI (two-tailed) $\approx (3.3, 10.7)$.
- Conclusion: Line A has higher mean output.